Downloads - main/src/main.c - Momentics IDE

File Edit Source Refactor Navigate Search Project Run Window Help

Run | bin.main | on: vm3

Project Explorer

- main
- raspb (connection lost, searching...)
- sahana
- vm2 (connection lost, searching...)
- vm3

qnx.c | main.c

Outline | Build Targets

- main
- sahana

Problems | Tasks | Console | Properties | Target File System Navigator

```
<terminated> bin.main [C/C++ QNX Application] /tmp/main on vm3 pid 1941506 (11/04/26, 9:50 am) (Terminated 11-Apr-2026, 9:50:39 am)
[INFO] [SYS]: Initializing QNX Framework-Driven RTOS...
[INFO] [SYS]: Enforcing SCHED_RR Real-Time Priority Policies...
[INFO] [CORE]: Dispatcher Framework Online. Priority 50 Set.
************************************************************
* SYSTEM ONLINE. Type 'q' and press ENTER to safely stop.   *
************************************************************

|   TEMP    | AI THRESHOLD | VOLTAGE | DENSITY | ALARM | COOLING SYSTEM |
|----------|--------------|---------|---------|-------|----------------|
| 30.00°C |    00.00°C  | 220.00V | 986.50  |   OK  |      IDLE      |
          Predicting failure. Engaging cooling subroutines.          Predicting failure. En
[AI ALARM] High Thermal Spiking Detected (+0.73°C/s)!
          Predicting failure. Engaging cooling subroutines.
| 32.46°C |    00.00°C  | 220.00V | 985.39  |  !!!  |     ACTIVE     |

[AI ALARM] High Thermal Spiking Detected (+0.69°C/s)!
          Predicting failure. Engaging cooling subroutines.
| 30.00°C |    00.00°C  | 220.00V | 986.50  |  !!!  |     ACTIVE     |

[AI ALARM] High Thermal Spiking Detected (+1.00°C/s)!
          Predicting failure. Engaging cooling subroutines.
| 30.00°C |    00.00°C  | 220.00V | 986.50  |  !!!  |     ACTIVE     |

[AI ALARM] High Thermal Spiking Detected (+0.94°C/s)!
          Predicting failure. Engaging cooling subroutines.
| 30.00°C |    00.00°C  | 220.00V | 986.50  |  !!!  |     ACTIVE     |

[AI ALARM] High Thermal Spiking Detected (+1.06°C/s)!
          Predicting failure. Engaging cooling subroutines.

[AI ALARM] High Thermal Spiking Detected (+1.42°C/s)!
          Predicting failure. Engaging cooling subroutines.

|   TEMP    | AI THRESHOLD | VOLTAGE | DENSITY | ALARM | COOLING SYSTEM |
|----------|--------------|---------|---------|-------|----------------|
| 30.00°C |    00.00°C  | 220.00V | 986.50  |  !!!  |     ACTIVE     |
```

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qnx.c | main.c

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <pthread.h>
5 #include <sched.h> // used for setting thread scheduling priority
6 #include <sys/neutrino.h>
7 #include <string.h>
8 #include <stdbool.h>
9 #include <time.h>
10
11 // SIMPLE LOGGING MACROS (for printing system messages)
12
13 #define LOG_LEVEL_INFO  "INFO"
14 #define LOG_LEVEL_CRIT  "CRIT"
15 #define SYS_LOG(level, module, msg) printf("[%s] [%s]: " msg "\n", level, module)
16
17
18 // STATES USED BY ACTUATOR (basic FSM)
19
20 typedef enum {
21     FSM_IDLE,
22     FSM_ENGAGING,
23     FSM_ACTIVE
24 } ActuatorState;
25
26
27 // THREAD PRIORITY VALUES
28 // higher number means higher priority in QNX scheduler
29
30 #define PRIO_SERVER      50
31 #define PRIO_AI          45
32 #define PRIO_ACTUATOR    40
```

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main

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qnx.c | main.c

```c
// CLIENT 3

void* client_actuator(void* arg) {
    usleep(100000);
    int coid = ConnectAttach(0, 0, server_chid, _NTO_SIDE_CHANNEL, 0);

    ActuatorState my_state = FSM_IDLE;
    Msg msg = { .msg_type = MSG_ACT, .fsm_state = my_state };
    Reply reply;

    while (1) {
        MsgSend(coid, &msg, sizeof(msg), &reply, sizeof(reply));
        if (reply.system_shutdown) break;

        switch(my_state) {
            case FSM_IDLE:
                if (reply.alarm_active) my_state = FSM_ENGAGING;
                break;
            case FSM_ENGAGING:
                usleep(200000);
                my_state = FSM_ACTIVE;
                break;
            case FSM_ACTIVE:
                if (!reply.alarm_active) my_state = FSM_IDLE;
                break;
        }

        msg.fsm_state = my_state;
        usleep(200000);
    }
    ConnectDetach(coid); return NULL;
```

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*********************************************************
```

main | 187M of 256M

27°C Mostly sunny | Search | ENG IN | 09:57 11-04-2026